

Homework-Game1: branching factor in games.

Suppose it is now "red" player's turn. See board situation on left figure.

(A) How many different/distinct moves he can make?

Write down details. For instance,

5个 兵 (soldier) has 5 moves

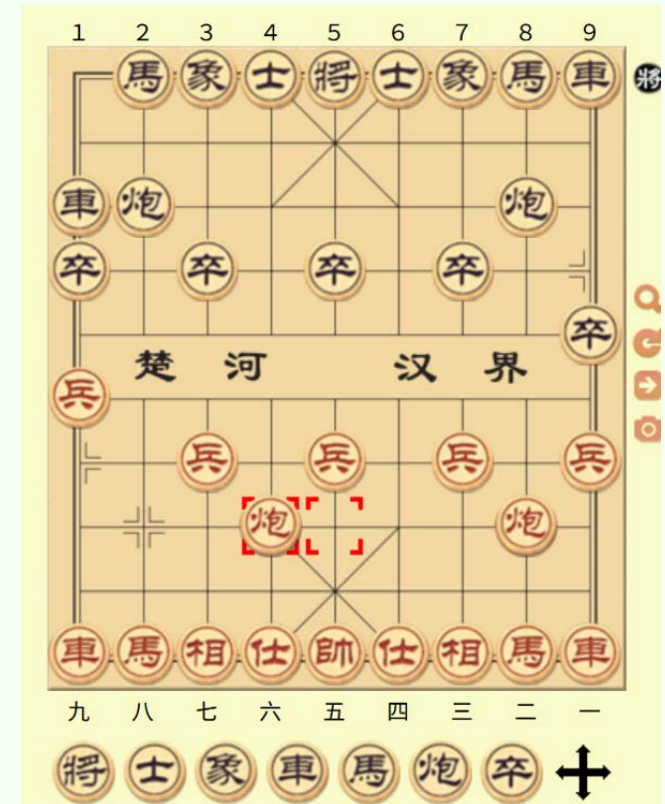
Left 炮 (cannon) has $3+3+1+6=13$ moves

...

This value is the branching factor, or number of adjacent nodes on the chess board situation graph.

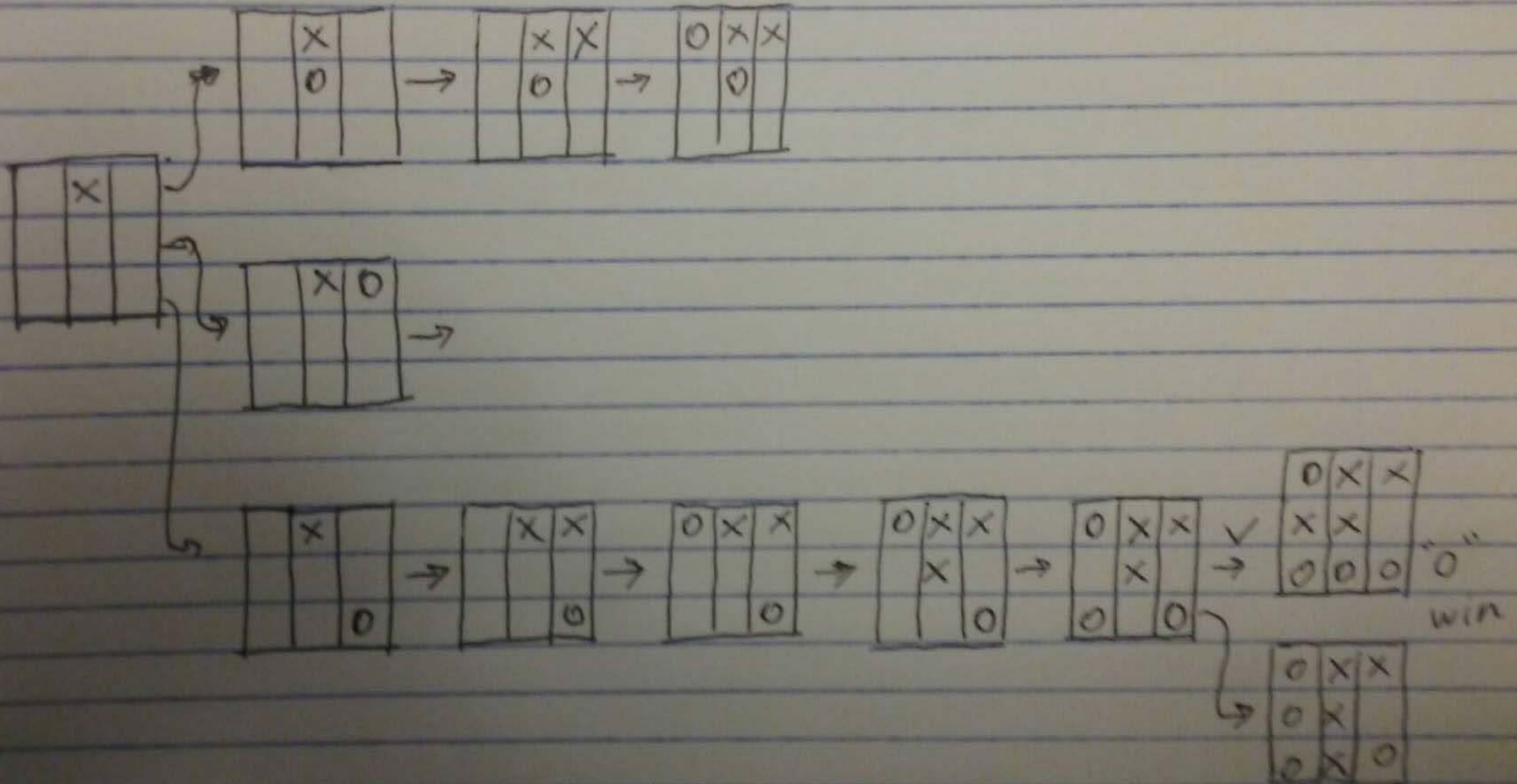
(B) If he use 1-step look-ahead strategy, how many evaluation of heuristic (assessment of the board situation) evaluation he needs to do?

(C) If he use 3-step look-ahead strategy, how many evaluation of heuristic evaluation he needs to do?



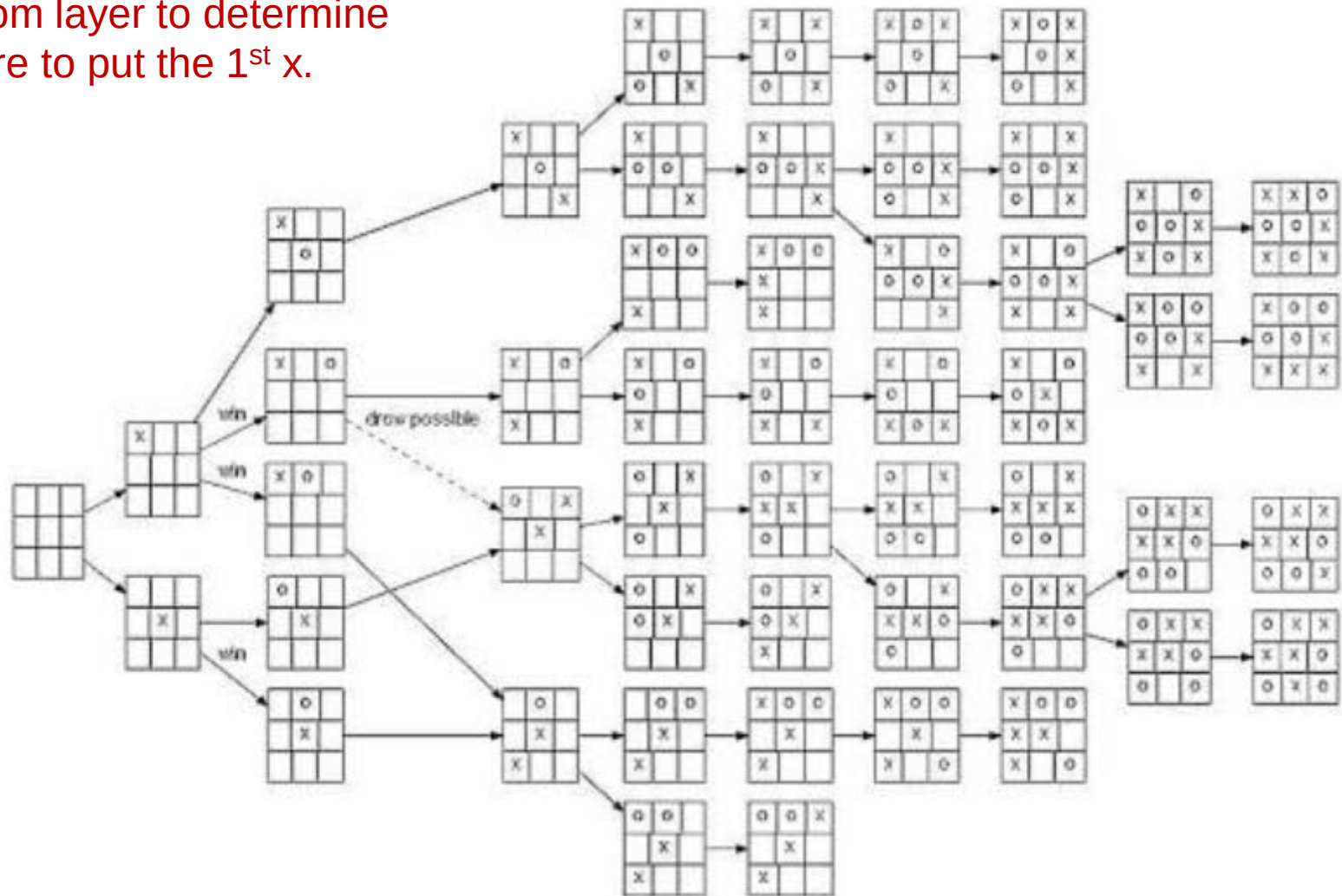
Homework-Game2: Complete the rest steps of the top two incomplete game plays using your own instincts.

Homework: You complete the rest/remaining games

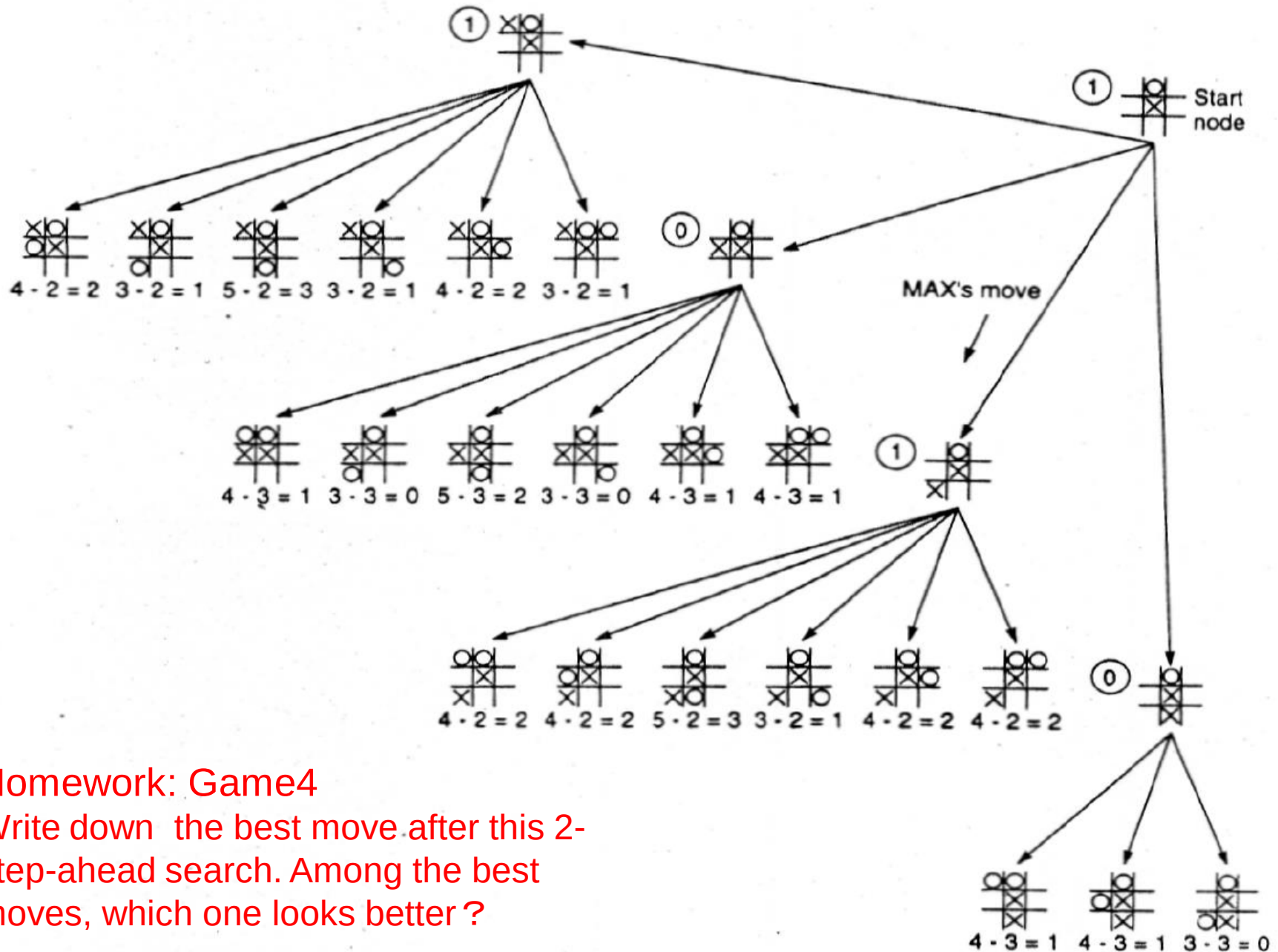


Homework: Game3:

Do value backup from the bottom layer to determine where to put the 1st x.

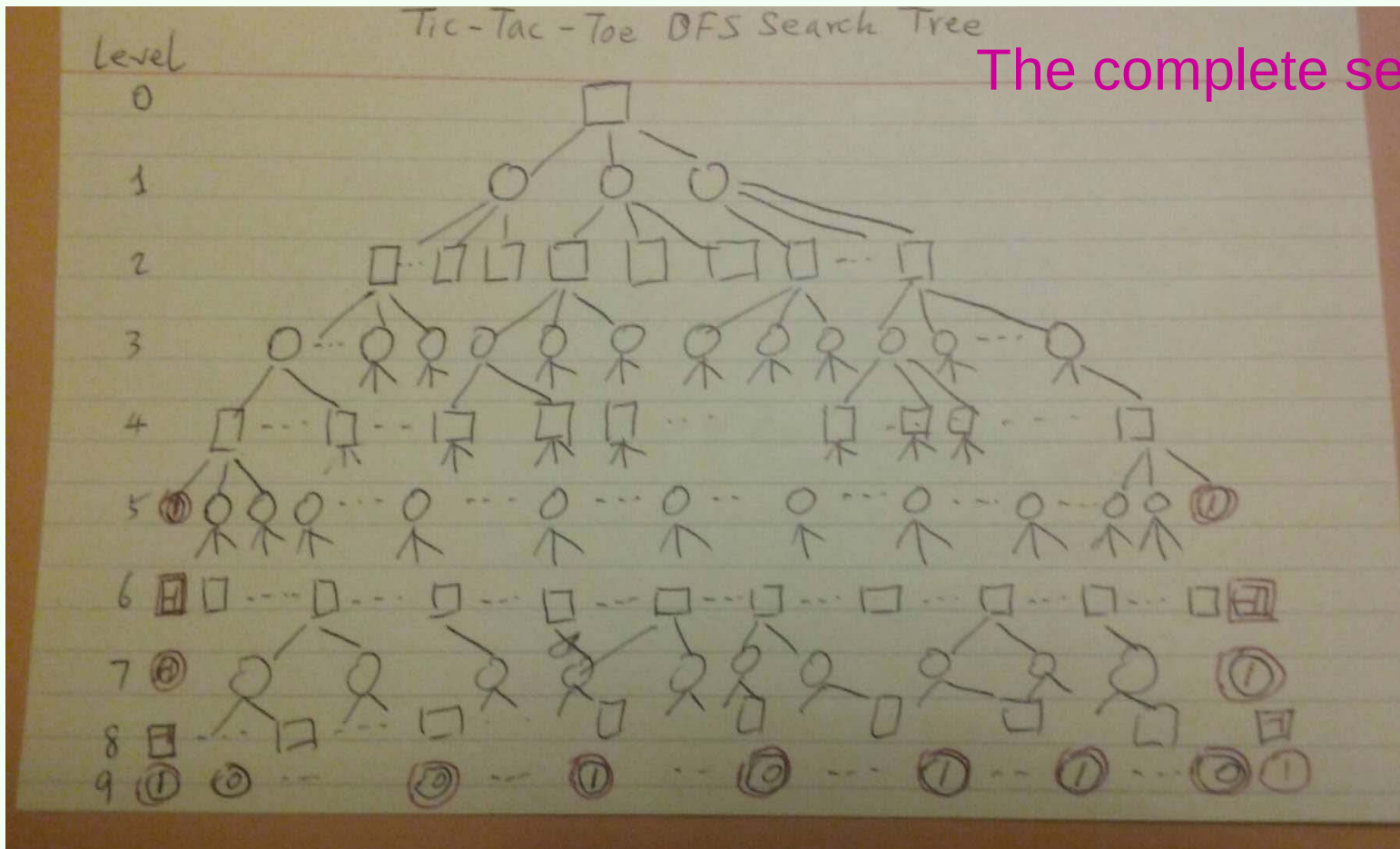


2-step-ahead search using heuristic evaluation function



Homework: Game4

Write down the best move after this 2-step-ahead search. Among the best moves, which one looks better?



Tree property:

1. Leaf nodes occur at levels 5,6,7,8,9. Why?
2. Leaf nodes on level 5 has value +1. Why?
3. Leaf nodes on level 6 have value -1. Why?
4. Leaf nodes on level 9 (bottom) have values 0 or 1. Why?
5. How many leaf nodes on level 5?
6. How many leaf nodes on level 6?
7. How many leaf nodes on level 9?

HOMEWORK-
Game5:
Explain these
properties

Homework: Game5. This tree is the same search tree in previous page

The complete search tree

